

Uber Data Analysis Project

Exploratory Data Analysis on Uber ride data — uncovering ride patterns, customer behavior, and operational trends.



The Questions We Set Out to Answer

Problem Statement

1

Top Category

Which ride category is booked most?

2

Purpose

Why do people book Uber rides?

3

Peak Time

When are cabs booked most?

4

Seasonality

Which months see fewer rides?

5

Day Trends

Which weekdays dominate?

6

Distance

How far do people typically ride?

Dataset Overview

START / END DATE

Ride start & end timestamps

CATEGORY

Business or Personal

START / STOP

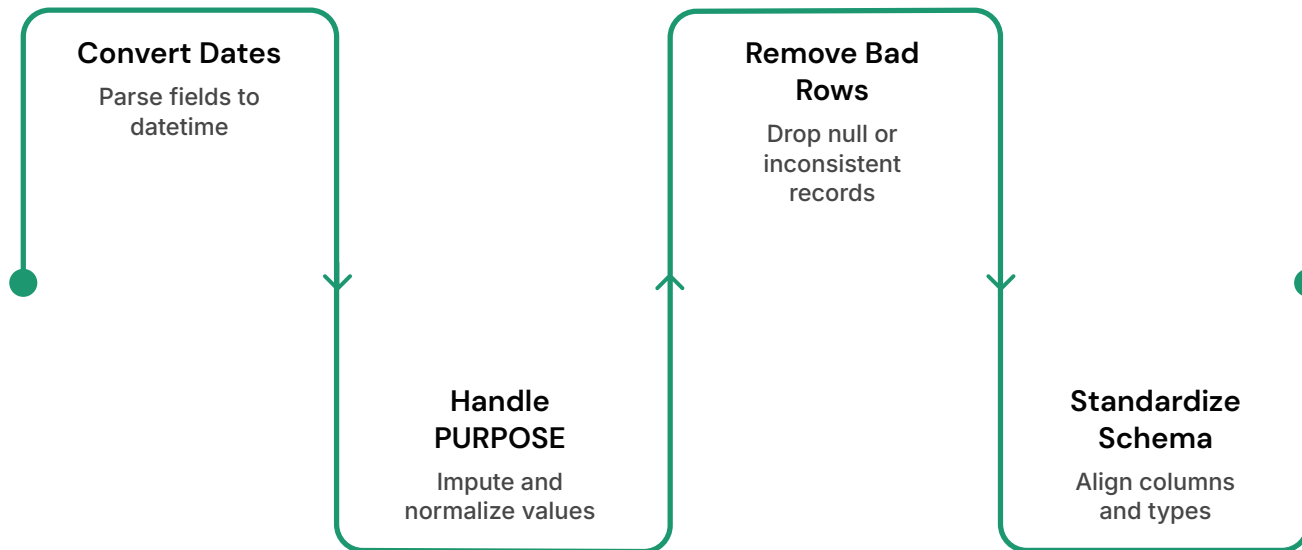
Pickup & drop locations

MILES & PURPOSE

Distance traveled & ride reason



Data Cleaning & Feature Engineering



New Features Created

Day

Monday – Sunday

Month

January – December

Hour

Extracted from start time

Time Category

Morning, Afternoon, Evening,
Night

Exploratory Data Analysis

Analysis Performed



Category Analysis

Business vs. Personal ride bookings



Time Analysis

Ride distribution across day periods



Day-wise Analysis

Weekday vs. weekend comparison



Purpose Analysis

Most common reasons for booking



Monthly Analysis

High & low ride frequency months



Distance Analysis

Typical trip length patterns

What the Data Revealed

Key Insights

 Business Dominates Business category has the highest ride count	 Work Purposes Meetings & work-related trips lead bookings	 Daytime Peak Morning & afternoon see highest demand
 Weekdays Win Weekdays far outpace weekends		 Short–Medium Trips Most rides cover moderate distances

Answers to Problem Statement

1 Category

Business has the highest number of rides

2 Purpose

Work-related (meetings) is the top reason

3 Peak Time

Daytime — morning & afternoon

4 Low Months

Certain months show seasonal dips

5 Top Days

Weekdays dominate over weekends

6 Distance

Most rides are short to medium distance

Business Use Cases



Demand Forecasting



Driver Allocation



Peak Time Identification



Pricing Strategy

Stack & Roadmap

Tools & Future Improvements

Tools Used

Python

Pandas, NumPy

Visualization

Matplotlib & Seaborn

Dashboard

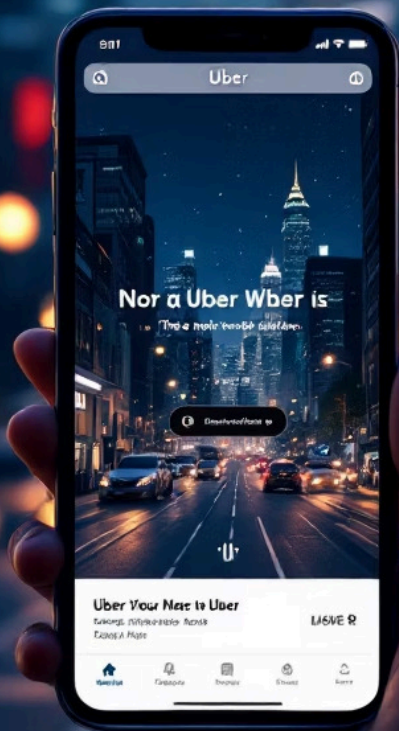
Power BI

Future Improvements

→ Location-based analysis

→ Predictive modeling

→ Ride duration analysis



Conclusion

Uber rides are primarily used for **business & work activities**, mostly **during daytime on weekdays**, and generally for **short to medium distances**.

This project demonstrates a complete EDA workflow — cleaning, feature engineering, and insight generation — supporting data-driven decision making.